Gudermannian function

1. give $\theta = gd(x) := \tan^{-1}(\sinh x)$

$\Rightarrow \sin\theta = \tanh x$, $x = \tanh^{-1}(\sin\theta)$

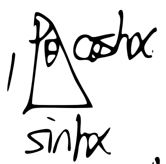
$\cos\theta = \mathrm{sech}\, x$

$\tan\theta = \sinh x$

proof: $\because \theta = gd(x) = \tan^{-1}(\sinh x)$

$\therefore \tan\theta = \sinh x$

draw



$\therefore \sin\theta = \frac{\mathrm{sh}\,x}{\mathrm{ch}\,x} = \tanh x$

$\therefore x = \tanh^{-1}\sin\theta$

$\therefore \cos\theta = \mathrm{sech}\, x$

2. Gudermannian: $gd(x): R \to (-\frac{\pi}{2}, \frac{\pi}{2})$

$gd(x) := \tan^{-1}(\sinh x)$

$\Rightarrow gd(x) \Leftrightarrow gd(x) := \int_0^x \mathrm{sech}(t)\,dt$

$\Leftrightarrow gd(x) := 2\tan^{-1}(e^x) - \frac{\pi}{2}$

proof:

① $\because (\tan^{-1}(\sinh x))'$

$= \frac{1}{\mathrm{sh}^2+1}\,\mathrm{ch} = \frac{\mathrm{ch}}{\mathrm{ch}^2} = \mathrm{sech}$

$\therefore \int \mathrm{sech}(t)\,dt = \tan^{-1}(\sinh t) + C$

$\therefore \int_0^x \mathrm{sech}\,t\,dt = \tan^{-1}(\sinh t)\Big|_0^x$

$= \tan^{-1}(\sinh x)$

② $\because (2\tan^{-1}e^x)' = \mathrm{sech}\,x$

$\therefore \int \mathrm{sech}\,t\,dt = 2\tan^{-1}e^t + C$

$\therefore \int_0^x \mathrm{sech}\,t\,dt = 2\tan^{-1}e^t\Big|_0^x$

$= 2\tan^{-1}e^x - \frac{\pi}{2}$

$\therefore gd(x) = \tan^{-1}(\sinh x)$

$= \int_0^x \mathrm{sech}\,t\,dt$

$= 2\tan^{-1}(e^x) - \frac{\pi}{2}$

3. inverse Gudermannian $gd^{-1}(\theta)$:

$(-\frac{\pi}{2}, \frac{\pi}{2}) \to R$

$gd^{-1}(\theta) = \int_0^\theta \sec(t)\,dt$ ①

$= \ln(\sec\theta + \tan\theta)$ ②

$= \ln\left(\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}\right)$ (2.2)

$= \ln\tan\left(\frac{\theta}{2} + \frac{\pi}{4}\right)$ (2.3)

$= \frac{1}{2}\ln\left(\frac{1+\sin\theta}{1-\sin\theta}\right)$ (2.4)

$= 2\tanh^{-1}(\tan\frac{\theta}{2})$ ③

$= \sinh^{-1}(\tan\theta) = \mathrm{sech}^{-1}(\cos\theta)$ ④

proof: ① $gd^{-1}(\theta) = \int_0^\theta \sec t\,dt$

$\because \theta = gd(x) = \int_0^x \mathrm{sech}(t)\,dt$

$\therefore x = gd^{-1}(\theta)$

$\frac{dx}{d\theta} = \frac{1}{\frac{d\theta}{dx}} = \frac{1}{\mathrm{sech}(x)} = \frac{1}{\cos\theta} = \sec\theta$

$\therefore \int \sec(t)\,dt = gd^{-1}(t) + C$

$\therefore \int_0^\theta \sec t\,dt = gd^{-1}(t)\Big|_0^\theta = gd^{-1}(\theta)$

($\because gd(0) = 0$, $\therefore gd^{-1}(0) = 0$)

(2.4) ② = (2.4), $\ln(\sec\theta+\tan\theta)=\frac{1}{2}\ln\left(\frac{1+s\theta}{1-s\theta}\right)$

proof: $\sec\theta+\tan\theta$

$$=\frac{1+s\theta}{c\theta}=\frac{1+s\theta}{\sqrt{1-s^2\theta}}=\sqrt{\frac{1+s\theta}{1-s\theta}}$$

∴ ② = (2.4)

(3) (2.2) = (3),

∵ $\tanh^{-1}(z)=\frac{1}{2}\ln\left(\frac{1+z}{1-z}\right)$

∴ $2\tanh^{-1}\left(\tan\frac{\theta}{2}\right)=\ln\left(\frac{1+t\frac{\theta}{2}}{1-t\frac{\theta}{2}}\right)$

(4): ∵ $\theta=gd(x)=\tan^{-1}(\sinh x)$

∴ $x=gd^{-1}(\theta)$

$=\sinh^{-1}(\tan\theta)$

$=\text{sech}^{-1}(\cos\theta)$ (from 1)

$=\tanh^{-1}(\sin\theta)$

(2) ① = ②, $\int_0^\theta \sec t\,dt=\ln(\sec\theta+\tan\theta)$

prof: ∵ $\theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

∴ $\sec\theta+\tan\theta=\frac{1}{c\theta}+\frac{s\theta}{c\theta}>0$

又∵ $\ln'(\sec\theta+\tan\theta)=\sec\theta$

∴ $\int \sec t\,dt=\ln(\sec t+\tan t)+C$

∴ $\int_0^\theta \sec t\,dt=\ln(\sec t+\tan t)\Big|_0^\theta$

$=\ln(\sec\theta+\tan\theta)$

(2.2) ② = (2.2), $\ln(\sec\theta+\tan\theta)=\ln\left(\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}\right)$

prof: ∵ $\sec\theta+\tan\theta=\frac{1+s\theta}{c\theta}$

又∵ $1+\sin\theta=\frac{\left(1+t\frac{\theta}{2}\right)^2}{1+t^2\frac{\theta}{2}}$

$c\theta=\frac{1-t^2\frac{\theta}{2}}{1+t^2\frac{\theta}{2}}$ (see 02062)

∴ $\frac{1+s\theta}{c\theta}=\frac{1+t\frac{\theta}{2}}{1-t\frac{\theta}{2}}$ proved.

(2.3) (2.2) = (2.3),

∵ $\tan\left(\frac{\theta}{2}+\frac{\pi}{4}\right)=\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}$ ∴ (2.2) = (2.3)